Multimedia Tools and Applications (2024) 83:65663–65685
https://doi.org/10.1007/s11042-023-17992-1

Quantum convolution neural network for multi-nutrient detection and stress identification in plant leaves

Kummari Venkatesh[1] · K. Jairam Naik[1] · Achyut Shankar[2]

Received: 17 October 2023 / Revised: 20 December 2023 / Accepted: 24 December 2023 /
Published online: 15 January 2024

Abstract

Nutrient stress can impose significant metabolic strain on plants, resulting in declining agricultural productivity. Nitrogen, phosphorus, and potassium are essential growth-limiting nutrients and are the primary building elements for amino acids, nucleic acids, proteins, and chlorophyll. The absence of these nutrients has observable effects on various plant characteristics, such as leaf size, color, and plant height. However, recent technological advances in imaging have given birth to computer vision-based plant phenomics, which holds great promise for plant research and management. The non-destructive, quick, automated assessments made possible by these imaging techniques are transforming the field of plant nutrient stress research and monitoring. This paper presents a hybrid quantum–classical model (HQCM) for identifying multi-nutrient stress and nutrient stress level quantification (NSLQ) for plant stress level identification by analyzing plant leaf images, utilizing the combined capabilities of classical and quantum computing systems. The HQCM model uses groundnut multi-nutrient stress (private), rice plant nutrient stress (public), and plant village (public) datasets for experimentation. The HQCM model exhibited impressive levels of accuracy, achieving rates of 97.79%, 97.37%, and 98.75% on the groundnut, rice, and plant village datasets, respectively. This performance surpasses conventional models, including Xception, DECM, DWC, and ResNet50V2. The proposed model showed significant advancements in accuracy, reaching the performance of state-of-art algorithms by 3.24%, 4.07%, and 6.73%, thus emphasizing its better performance.

Keywords Multi-nutrient stress · Plant leaves · Quantum computing · Nutrient stress level detection

✉ K. Jairam Naik
Jnaik.cse@nitrr.ac.in

Kummari Venkatesh
kvenkatesh.phd2020.cse@nitrr.ac.in

Achyut Shankar
ashankar2711@gmail.com

[1] Department of Computer Science and Engineering, National Institute of Technology Raipur, Raipur, India

[2] WMG, University of Warwick, Coventry, UK

1 Introduction

Quantum machine learning (QML) epitomizes a burgeoning field that seamlessly integrates the foundational tenets of quantum physics with standard machine learning practices [1, 2]. Within this realm, there lies the promise to transform various facets of computing, with a notable impact on image classification [3, 4]. It has attracted researchers because it efficiently handles the computational problems in conventional computers. This potential originates from the distinctive characteristics of quantum computation, namely entanglement and superposition, which can expedite some ML tasks significantly using exponential acceleration [5]. In addition to that, QML algorithms can produce probabilistic results, making them well-suited for addressing classification problems [6]. Furthermore, these entities function within a search area that exhibits exponential growth, substantially improving their overall performance [7–9]. Nonetheless, the integration of quantum algorithms into practical applications presents significant challenges, including the crucial need for error rectification and the heightened susceptibility of quantum frameworks to external disruptions [10]. Despite these obstacles, quantum machine learning (QML) has showcased promising results in various applications [11]. QML algorithms have shown superior efficiency in handling large image datasets, particularly in the area of image classification, as compared to conventional methods. This efficiency translates into faster and more accurate classification results [12].

In recent years, increasing research has explored the correlation between ML and quantum technology (QT). They have investigated whether QT can improve conventional ML and deep learning (DL) techniques, such as dimension reduction and supervised learning [13, 14]. Cong et al. [15] devised a tailored convolutional neural network (CNN) based on quantum circuits, demonstrating precise classification of quantum states. Kossaif and Bulat [16] employed a parameterization technique involving a single higher-order tensor in their global CNN. Tensor methods are an efficient way to represent network structures. Incorporating a low-rank framework into this tensor serves a dual purpose: it both regularizes the network and reduces the parameter count, resulting in improved accuracy. The Henderson et al. [17] was developed as a quantum convolutional layer for feature extraction in image classification. This layer utilizes the random quantum circuit to achieve this objective. The software framework developed by Broughton and Verdon [18] presents a method for QML. This approach involves the construction of quantum circuits for supervised classification by assembling a succession of quantum gates. The utilization of quantum circuits is essential within ML, as emphasized in [19–21]. Li et al. [22] introduced the hybrid quantum–classical processing paradigm and variation quantum computation methods. They deployed and validated this framework on the MNIST dataset, demonstrating its viability and superiority over CNNs.

This research article presents the HQCM model for multi-nutrient stress identification using plant leaf images and determined nutrient stress levels in the groundnut crop. The acquisition of plant leaf images takes place in groundnut crop fields utilizing a Raspberry Pi device equipped with a camera module connected to it. Furthermore, the images are categorized according to the symptoms exhibited by plant leaves, and image preprocessing techniques are utilized to remove any unwanted noise. The generated images are subsequently inputted into the HQCM model to detect multi-nutrient deficits. Later, the images depicting manifestations of multi-nutrient stress are subjected to additional scrutiny to ascertain the existence of multi-nutrient stress inside the plant's leaves. The key contributions outlined in this research article include the following.

1. Real-time groundnut crop leaf images are collected using the Raspberry Pi device equipped with a camera module connected to it.
2. We proposed the HQCM for multi-nutrient stress identification in the groundnut plant. This model utilises the advantages of quantum and classical computing principles. The quantum circuit is used for feature extraction in the initial layers of the HQCM. Later, classical CNN identifies the multi-nutrient stress in the plants.
3. The categorized images are further examined to determine the nutritional stress levels. The assessment involves an analysis of the green colour intensity and the probability distribution function (PDF) of the image.

The subsequent sections of this research article are organized as follows: Sect. 2 provides an in-depth review of the existing literature. Section 3 presents a comprehensive elucidation of the proposed methodology employed in this study. Section 4 entails the analysis of results. Finally, Sect. 5 deliberates on the study's conclusion and delineates potential avenues for future enhancements.

2 Literature survey

In noisy intermediate-scale quantum (NISQ) computing [23], a substantial proportion of information processing activities heavily depend on traditional computing. Quantum computing methods are presently employed to increase the efficiency of classical ML and DL algorithms, serving as essential elements in enhancing information processing capacities [24]. Classification is a common objective in both ML and DL methodologies. In contrast, quantum computing functions by changing data in a superposition state via the exploitation of entanglement. It is worth noting that quantum computing exhibits sampling and probabilistic computing properties similar to traditional ML techniques. This similarity allows the application of QML methods to classical image processing approaches [25–27]. Figure 1 presents a brief overview of research endeavors on quantum image classification techniques.

The CNN-based DL algorithms are widely prevalent in plant science to identify nutrient deficiencies, weed identification and plant disease by analyzing plant leaf images. The increasing density of weed plants in fields of crops presents a significantly heightened risk to crop plants due to competition for depletion of key nutrients. The authors have devised a system for identifying weeds, which is further discussed in [28]. In developing this system, the authors utilized RGB images and deployed the

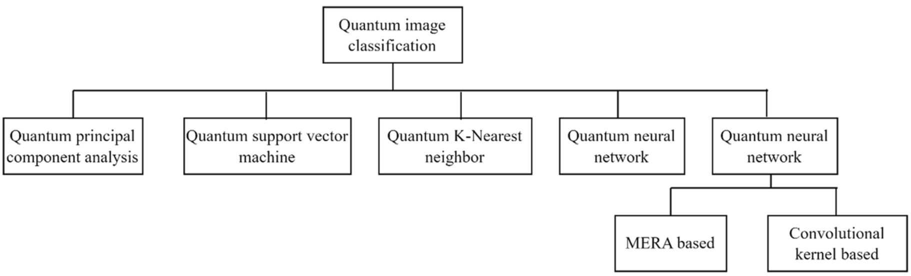


Fig. 1 Image classification methods using quantum computing [6]

pre-trained VGG16 architectural model. Significantly, this model has exhibited a noteworthy level of accuracy, reaching 95%, by leveraging a collection of 70 distinct features. The authors of [29] provide a novel image classification system that utilizes transfer learning. This system incorporates a pre-trained neural network based on Xception architecture for extracting features. The feature extraction process involves using depth-wise and point-wise convolution operations. The model above presents the benefit of accelerated training, though limited by its complex architectural design. A technique for identifying physiological disorders in apple plants utilizing the ResNet50V2 [30] model is introduced. The method mentioned above demonstrates a categorization accuracy of 94%. One of the limitations of this architecture pertains to the utilization of skip connections.

The emergence of variational quantum methods has been a key research topic throughout the NISQ computing era, as detailed in [31]. Quantum Neural Networks (QNNs), utilizing quantum circuits, have garnered substantial attention from the scientific community, standing out among the algorithms currently under scrutiny. These networks consist of an assembly of quantum gates, with a central focus on fine-tuning the parameters linked to these gates to attain specific goals. HQ algorithms have proved noticeable advantages when collated to their traditional counterparts. In 2018, Farhi et al. [32] presented a QNN that incorporated the optimization of quantum circuit parameters to perform binary classification tasks and achieved 97% classification accuracy. In the work mentioned above, Schuld et al. [33] have made advancements in the architecture of neural networks by introducing a quantum classifier centered around circuits, hence streamlining the construction of QNNs. Furthermore, Pesah et al. [34] have shown that by constructing meticulously built quantum circuits, quantum neural networks (QNNs) may overcome the challenges posed by gradient descent, which are commonly seen in classical neural networks. This observation underscored the inherent potential benefits of Quantum Neural Networks (QNNs).

The Qin [35] et al. introduced graph-based image analysis method in HiGraph+, a smart computer learning tool for videos without labels. It uses a hidden graph network to understand how things change over time. By comparing different parts of the video, it learns to recognize patterns. Our tool works well for different tasks, but sometimes it struggles with big changes or messy backgrounds in videos. Yan [36] et al. introduces a new way, VL-Prompt (VLP), for making computers describe videos better. We use special phrases in two steps: first, to teach the computer about videos, and then to help it describe them. One issue is it might not work well in some situations due to limited data. Qin [37] et al. present HEVis*, an advanced video segmentation model surpassing current methods. Employing innovative techniques, HEVis* excels in comprehending visual dynamics, predicting diverse objects in videos efficiently. Rigorous testing establishes its effectiveness, setting new benchmarks in video analysis and demonstrating superior object understanding and representation capabilities. Qin [38] et al. presents VideoHiGraph, a self-supervised learning method for comprehending temporal relationships in unlabeled videos. Leveraging consistent patches and a graph kernel, it trains a hidden graph network for temporal coherence, enhancing representation learning and correspondence inference. Liu [39] et al. introduces TFPN (Tripartite feature pyramid network) to help computers understand detailed images better, especially for tasks like finding objects. TFPN is an improved version compared to basic FPN, using three smart ideas: focusing on important details, aligning features perfectly, and having a conversation between different computer layers.

Our study involved a thorough review of many research papers on the approaches used for image classification in both quantum and classical environments. Several significant research gaps have been identified during this study, which will be explained as follows.

1. Existing models can detect individual nutrient stress related to nitrogen, phosphorus, or potassium in plants. However, these models cannot identify multiple nutrient stress combinations involving nitrogen-phosphorus, phosphorus-potassium, or potassium-nitrogen.
2. Existing models cannot accurately detect the levels of multi-nutrient nutritional stress in plants.

3 Proposed methodology

This section explains the proposed method and its components, which are components of HQCM and NSLQ. The HQCM module detects the multi-nutrient stress in the plants, whereas the NSLQ module is employed to quantify the nutrient stress levels.

The HQCM utilized traditional and quantum computing technologies. Quantum computing uses fundamental principles and distinctive characteristics of quantum physics to execute computational tasks, including quantum bits (qubits), entanglement, superposition, and interference. Quantum computing has the potential to provide more efficient solutions to complex problems than standard computing technologies [40, 41]. In the classical computing counterparts, qubits function as the basic entities for information manipulation in quantum computing systems. In contrast to traditional bits, qubits can simultaneously occupy multiple states, such as the zero-state, one-state, or a combination of both states, referred to as linear superposition. The qubits are denoted as state vectors residing in Hilbert space [42].

$$\langle\psi\rangle = \begin{pmatrix} \theta \\ \delta \end{pmatrix} = \theta\langle 0\rangle + \delta\langle 1\rangle \tag{1}$$

The probability amplitudes δ and θ are denoted by complex numbers and $\theta^2 + \delta^2 = 1$

In accordance with the principles of quantum mechanics, every quantum gate (QG) can be characterized as a unitary transformation, which is effectively represented by a unitary matrix. To ensure that a matrix is unitary, it must satisfy the following conditions:

$$JJ^{+} = J^{+}J = I \tag{2}$$

Here J^{+} represents the complex conjugate of matrix J and I represent the identity matrix.

3.1 Quantum convolutional layer

The proposed method (HQCM) uses the quantum convolutional (quanvolution) layer, is represented in Fig. 2, which integrates the concept of quantum convolution as elucidated by Henderson et al. [17]. The concept of quantum convolution entails the utilization of small, randomized quantum circuits (RQCs) to execute convolution operations. This approach is specifically devised to ensure compatibility with quantum hardware. This methodology involves the application of the region-based quality control (RQC) method to distinguish localized regions within input images, intending to extract essential and useful features. The model integrates a quanvolution layer, which comprises three fundamental phases: encoding, a random quantum circuit (QC), and decoding.

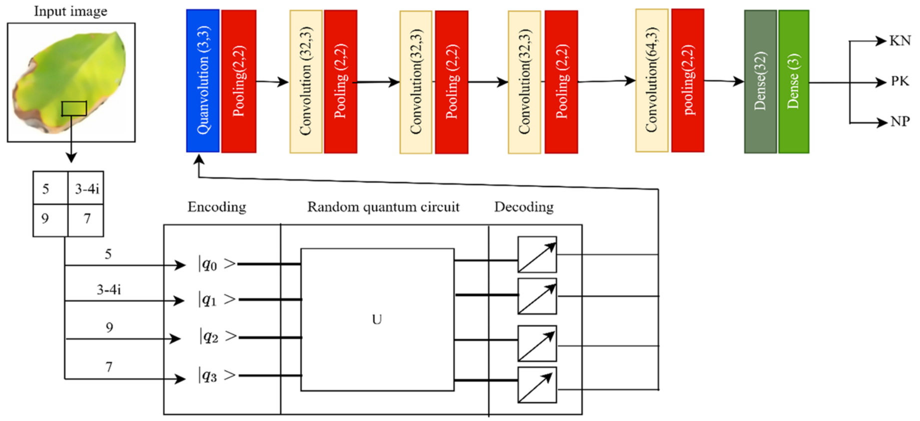


Fig. 2 Proposed HQCM model for multi-nutrient stress identification

3.1.1 Encoding

The encoding process entails transforming classical data into quantum data. The encoding circuit takes the input data, denoted as X belonging to the real-number space R^n, and maps it into a 2^n dimensional Hilbert space known as $\mathcal{H}$. This mapping is accomplished using a quantum circuit denoted as $E(X)$ which consists of n qubits, and it is facilitated by a quantum encoding function called $\phi_Q(x)$. This entire process is carried out in conjunction with a unitary circuit referred to as $U(X, \theta)$, as described in Eq. (3).

$$E : X \mapsto U_{\phi Q}(X)|0\rangle\theta^n \tag{3}$$

In the quantum circuit $E(X)$, the initial step involves subjecting every input element i from the set X to the threshold technique. This threshold technique creates a superposition state represented by $(\psi\rangle$ and converts the classical data to quantum data using the threshold technique. If value is greater than 120 it becomes the 1 otherwise zero. This superposition states are further modified through the randomized circuit.

3.1.2 Randomized quantum circuit

A QC is characterized as a sequential arrangement of quantum unitary operations, typically referred to as gates, and measurement procedures. These components are interconnected by qubits or quantum wires. The quantum convolutional layer, identical to its classical counterpart, consists of quantum kernels that are utilized to process an input image. The fundamental principle underlying quantum convolution entails the utilization of random quantum circuits to partition the input image into separated, localized sections. This partitioning enables the quantum circuit to extract significant and interpretable characteristics from these confined segments selectively, matching how classical convolutional layers discern features across distinct regions of an image. The involvement of this RQC is essential in transforming conventional data into quantum states. It actively explores diverse quantum feature spaces during the training process, thereby contributing to data augmentation techniques. Specifically designed for feature extraction in images, the RQC is instrumental in enhancing the model's capacity for nuanced representation.

Filter generation The initial step in constructing each quantum filter involves determining the dimensions of the quanvolutional filter. This parameter denotes the quantum mechanical units, known as qubits, that are necessary for the execution of the circuit. This research employed a straightforward methodology by utilizing 3×3 quanvolutional filters. Consequently, each simulated circuit consisted of 9 qubits, denoted as $n = 3$. The subsequent step involved the construction of the physical circuit, wherein each qubit was regarded as an individual node within a graph. A probability of connection was assigned to each pair of qubits. The probability denotes the likelihood of implementing a 2-qubit gate between the abovementioned qubits. The gate selection process was conducted randomly and encompassed a range of gate types, such as Swap, CNOT, $\sqrt{SWAP}$, and Controlled U gates. Moreover, a set of 1-qubit gates, including $X(\theta), Y(\theta), U(\theta), Z(\theta)$, T, H, and P (where θ represents a randomly chosen rotational parameter), were produced randomly. The selection of the target qubit for each of these gates was performed randomly.

3.1.3 Decoding

The decoding process, also known as the measurement stage, is pivotal in the transformation of quantum data into a conventional format. A frequently employed methodology for measuring entails the utilization of Pauli matrices, as referenced in [32]. The decoding step in the HQCM model utilizes the threshold technique, as mentioned in the encoding section.

3.2 HQCM model for multi-nutrient stress detection in plants

This section explains the proposed Hybrid Quantum–Classical Model (HQCM), which integrates classical and quantum computing technologies. This HQCM model introduces quantum technology into the proposed methodology; due to this, some additional advantages are included, such as parallelism, Quantum Fourier transform (QFT), Handling large datasets, Feature extraction, and Improved optimization. Figure 2 shows the proposed model architecture diagram. This diagram comprised of quanvolution, convolution, and polling layers. Finally, Dense layers are used to classify the multi-nutrient stress in the plants. In conventional deep learning, the feature extraction model utilizes convolutional and pooling layers to obtain significant and non-repetitive features from image data denoted as $U = \left[u_1, u_2, u_3, \ldots\ldots, u_N\right]$. The standard convolutional layer (C_x) computes feature values, represented as $U_{ij}^{C_x}$, by examining the pixel values inside the domain U. During this procedure, a conventional filter systematically scans over local areas located at the (p, q) location of the image. It determines the dot product by multiplying the mask with the pixel values of the image within the local area. The weights, denoted as W_{ab} are obtained from a conventional convolutional filter with dimensions $F_c x F_c$ as specified in Eq. (4), where F_c is the size of the convolutional kernel.

$$U_{ij}^{lc} = \sum\nolimits_{a,b=1}^{F_c} W_{a,b} U_{p+a-F_c/2,q+b-F_c/2}^{lc-1} + bias \tag{4}$$

The features extracted are then mapped onto the conventional feature space (F) through the traditional feature mapping function ϕ_c, denoted as ϕ_c: U → F. Pooling operations are employed to shrink the spatiality of the feature map, while dense layers flatten the data and consolidate information gathered from previous layers, preparing it for the classification task. Throughout training, the filter weights undergo continuous updates and fine-tuning to attain optimal performance.

Let the quantum kernel size be denoted as F_c. The randomized quantum circuit can be organized as a QCNN layer in quantum computing. The QCNN layer consists of quantum filters designed to execute a convolution inner product operation between U and F, where both matrices have dimensions of $F_c x F_c$. The execution of this operation is carried out via a quantum filter circuit, as explained in [28].

The $E(x)$ performs the function ϕ_Q of quantum feature map encoding, translating the input image data U to the Hilbert space $\mathcal{H}$. The encoding procedure entails translating the pixel values in U into quantum states denoted as $\phi_Q(x)$ that exist in Hilbert space, represented as $\phi_Q : x \mapsto |\phi_Q(\mathrm{x})\rangle$. Conversely, the randomized circuit obtains information about the image characteristics. This is accomplished by implementing unitary gate operations and repeated quantum state measurement in $|\phi_Q(\mathrm{x})\rangle$. The subsequent values are transmitted to the traditional CNN architecture for additional feature extraction at more advanced stages. Traditional CNN filters, each with a size of 3×3 and a total of 32 filters, are employed for feature extraction. The features that have been extracted undergo dimensionality reduction via the application of the max-pooling operation with a dimension of (2,2). This iterative process continues until reaching the dense layer. The final two layers are characterized by dense layers containing 32 and 3 neurons, respectively. The last three neurons are specifically utilized for identifying multi-nutrient deficiencies in plant leaves, achieved through the application of the SoftMax function.

3.3 NSLQ model for determination of multi-nutrient stress levels in plant leaves

This section explains the multi-nutrient stress level identification in the plant leaves. In a real-time environment, plants may experience concurrent nutrient stresses. To detect this multi-nutrient stress, we employed the NSLQ method specifically designed for assessing such conditions in plants. Identification of multi-nutrient stress can be accomplished by analyzing the color pigmentation of plant leaves, where the green color serves as an indicator of the plant's overall health. The Green Index (GI) is employed to assess the levels of multi-nutrient stress in plant leaves because green coloring indicates the plants' health status. It is essential to acknowledge that a clear relationship exists between the GI value and the Chlorophyll Meter Index (CMI) value [43]. This CMI value represents the nutrient stress levels in the plant leaves. Figure 3 illustrates the procedure for determining the green index (GI) value.

The HQCM model is utilized to identify the presence of multi-nutrient stress in plant leaves. The aforementioned model demonstrates an accurate classification of the various nutrients that exist in plant leaves, comprising NP, PK, and KN. After the process of classification, the images are organized and kept in three distinct folders, which facilitates further in-depth study. The mentioned folders are systematically separated into ten discrete portions, whereby the nutrient stress level quantification (NSLQ) approach is carefully implemented. The images undergo systematic processing using the NSLQ method within the allocated folders in order to generate the GI value.

Determining the GI value involves a series of consecutive procedures. In the first step, plant leaf images are subjected to split, which divides them into distinct color channels: red, green, and blue. Each color channel is then processed individually. With a specific emphasis on the green channel, we compute the mean value and subsequently determine the range of minimum and maximum pixel values based on this mean value. For example, if the mean value of the green channel is represented as X, the initial range of pixel values for this channel is assigned as X. Subsequently, the probability (P) of encountering X inside

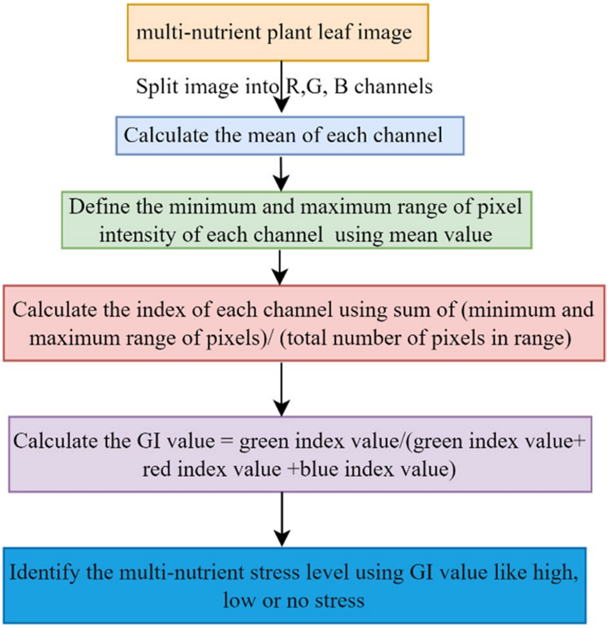


Fig. 3 GI value calculation model

the green channel is computed. In instances when the calculated probability (P) is less than 0.9, we gradually increase the range of intensity values, including both the minimum and maximum, by one unit. The iterative process persists until the P attains the predetermined threshold of 0.9. The primary justification for implementing this approach is its capacity to efficiently eliminate extremely low and high-intensity values in the image channel, hence enhancing the precision of the GI as an indicator of plant health. Once the P value attains the threshold of 0.9, the procedure halts any further increments of the minimum and maximum range. Subsequently, the Green Channel Value (GCV) calculation is as follows: it involves summing up the total pixel intensity values and dividing this sum by the quantity of pixel values within the channel. Similarly, this identical procedure is iterated for the blue and red channels within the image. The GI value computation, which relies on the values extracted from the image's green, red, and blue channel values, is performed according to the formula depicted in Eq. (5).

$$GI = \frac{GCV}{GCV + RCV + BCV} \tag{5}$$

where GCV, RCV, and BGV represent the green, red, and blue channel values, respectively. A GI score below 0.95 denotes a low quantity of nutrient stress, but a GI value above 1.05 signifies a severe level of nutrient stress in the plant leaves. On the contrary, noteworthy nutritional stress is absent when the GI score is between 0.95 and 1.05.

The comprehensive design of the proposed model is visually represented in the flowchart diagram presented in Fig. 4. The proposed model comprises two modules, namely HQCM and NSLQ. The HQCM module is utilized to identify the multi-nutrients in plants. In contrast, the NSLQ module is specifically engineered to evaluate the stress levels caused by these nutrients in plants. In the first stage, data is collected from the crop field using a Raspberry Pi and image pre-processing techniques are employed. This pre-processed dataset is segregated into two sets: testing and datasets. These datasets are used for training and

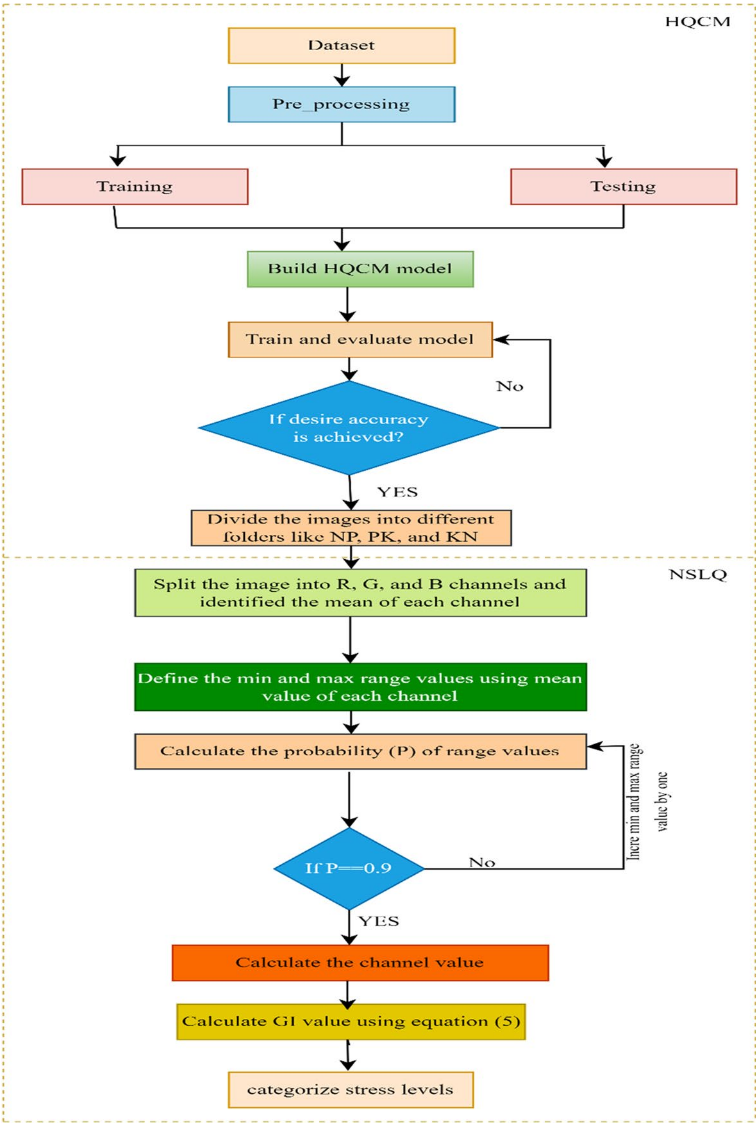


Fig. 4 Proposed model flow chart diagram

assessing the HQCM model. The HQCM model detects multi-nutrients in the plants. Once the desired accuracy is achieved, the images are systematically classified and organized into distinct folders corresponding to the nutrient stress categories of NP, PK, and KN. If the intended level of accuracy is not achieved, the iterative training procedure continues until the target accuracy is eventually accomplished. The categorized images are passed to the NSLQ model for further analysis.

The NSLQ model initially divides the RGB image into its separate red, green, and blue channels, subsequently calculating the mean intensity for each channel. Later, the algorithm determines the minimum and higher bounds of intensity values in the image channel by utilizing the mean value. It then proceeds to compute the probability of pixel values falling within this specific range of intensities. If the calculated probability value is less than 0.9, the intensity value range is raised by one iteratively. The procedure above persists until the probability value attains the threshold of 0.9. Once the probability value exceeds the threshold of 0.9, the determination of each channel's value entails dividing the sum of all pixels in the image channel by the total pixel count. The GI value is subsequently calculated using Eq. (5). The GI values are utilized to categorize the multi-nutrient stress levels in the plant.

4 Results analysis

The numerical outcomes were derived using the computing infrastructure parameters provided in Table 1. As stated by the authors in [44], the Pennylane framework provides users with the capability of modelling quantum computational units (QPUs) at several stages, including the design, training, and evaluation of hybrid quantum–classical (HQC) models. Following this, during the testing process, these models' quantum components can be implemented on a physical QPU. Therefore, the proposed HQC model can operate on a classical computer using QPU simulation.

4.1 Data acquisition

This section provides an extensive summary of the data collection methodology employed in the present study. In the present study, three separate datasets were utilized for experimental purposes: the multi-nutrient groundnut leaf image dataset [45], the rice plant dataset [47], and the plant village dataset [46]. The images of groundnut plant leaves were acquired in the agricultural fields of the Kalwakurthy municipal corporation in the Nagarkurnool district of Telangana, India. These plant leaf images are acquired using a Raspberry Pi equipped with a mounted camera module, having the following specifications: Model B containing, 4 GB RAM, quad-core CPU with 64-bit architecture, operating at frequency of 1.5 GHz and aperture size – f/2.9; focal length—3.29 mm; resolution -1080 pixels; FOV—72.4°. The setup for data collection is depicted in Fig. 5. The collected data was validated using [48, 53, 54], and comprehensive information about datasets is shown in Table 2.

Table 1 Computational resources

Part	Item	Specifications
Hardware	CPU RAM	Intel core i9 @2.80 GHz 32 GB RAM,
Software	System Programming Language Backend	Windows 10 pro OS Python 3 Pennylane, TensorFlow, Kera's

Fig. 5 Image acquisition setup

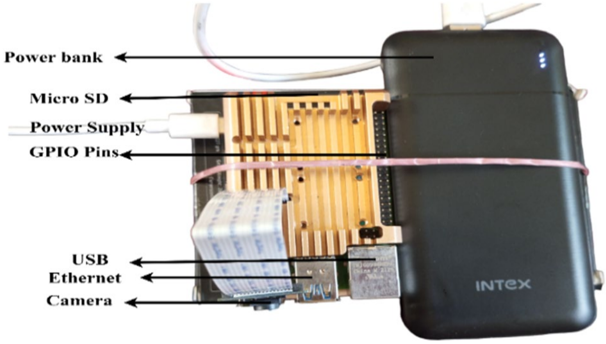


Table 2 Dataset details

Name	Type	Class	Number of images
Groundnut plant	Private	NP	1598
		PK	1734
		KN	1815
Rice plant	Public	NP	1238
		PK	1233
		KN	1177
Plant village	Public	Apple_scab	1621
		Apple_blackrot	1794
		Apple_cedarapplerust	1488

4.2 Image preprocessing

This section presents a comprehensive elucidation of the methods employed for pre-processing data. In this section, a range of image pre-processing techniques has been utilized to enhance the overall quality of captured images and alleviate the impact of environmental parameters on image quality. The mean, median, and Gaussian filters are opted for noise removal techniques in the pre-processing. The fundamental principle underlying mean filtering is relatively simple: it entails replacing the value of every pixel in an image with the mean of its surrounding pixels, which includes the pixel itself. The procedure above effectively removes pixel values that do not precisely represent their immediate surroundings. The median filter principle remains the same as a mean filter but replaces the median instead of the mean. The Gaussian filter smoothed the image by applying a weighted average to individual pixels. These weights are derived based on the Gaussian distribution. This procedure mitigates the noise and improves the level of detail by minimizing the influence of distant pixels on the present pixel. The mathematical notation of mean, median, and gaussian filters is given in Eqs. (6) to (8), respectively.

$$m'(p,q) = \frac{1}{rs}\sum\nolimits_{(s,t)\varepsilon S_{pq}} g(s,t) \tag{6}$$

where S_{pq} represent the collection of coordinates within the sub image, where the filter size is specified as (r, s). Additionally, let $g(s, t)$ reflect the mean value of the distorted image within the region defined by S_{pq}.

$$M'(p,q) = median_{(s,t)\varepsilon S_{pq}}\{g(s,t)\} \tag{7}$$

where $g(s,t)$ denotes the input image and S_{pq} represents the box size sub-image with filter dimension *pxq* which is centered at *p, q*.

$$G(p,q) = \frac{1}{2\prod\sigma^2} e^{\frac{-\left(p^2+q^2\right)^2}{2\sigma^2}} \tag{8}$$

Here the variables *p* and *q* represent the pixel positions in the image. σ signifies the standard deviation. The visual appearance of images remains unchanged compared to their post-preprocessing counterparts when pre-processing is omitted, but the modifications within the images are outlined in Table 3. The data in the first three columns of this table pertain to the input image parameters, explicitly denoting the mean (*m*), standard deviation (SD), and green Index (GI) values. The green Index value is determined by utilizing pixel values within the image. In essence, a color image is comprised of RGB channels. The GI value is computed as the *m* value of the green channel divided by the *m* value of the combined RGB channels. This GI value serves as an indicator of the plant's healthiness. The PSNR is estimated after applying the noise removal filter on the pre-processed image because PSNR represents the quality of the image.

Furthermore, we employed image augmentation and normalization techniques for images. The primary objective of image augmentation is to expand the dataset's size without additional data acquisition, enhancing the model's capacity for generalization. In DL, the normalization process plays a significant role in facilitating the convergence of models. The changes in the green channel of the initial image after the implementation of image pre-processing can be observed in Table 4. The table showcases various aspects of image processing. The initial row exhibits the input image, while the second row depicts the histogram of the green channel image. The subsequent rows, namely rows 3, 4, and 5, demonstrate the alterations observed in the histogram of the green channel after applying mean, median, and Gaussian filter pre-processing techniques, respectively.

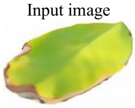
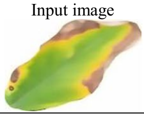
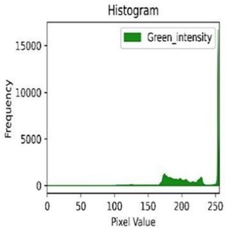
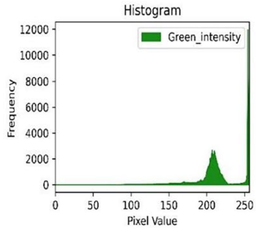
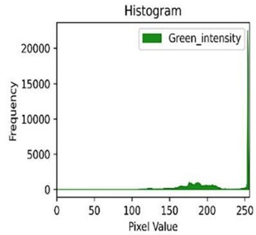
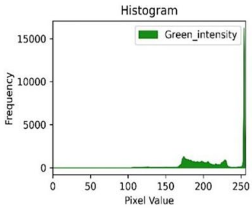
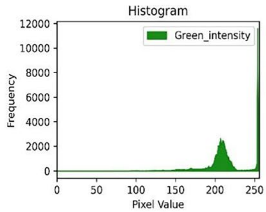
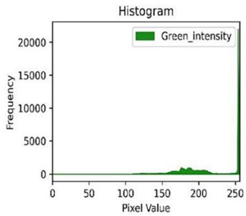
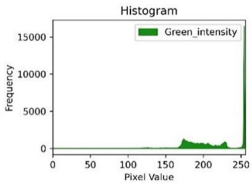
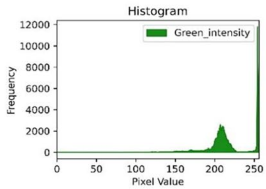
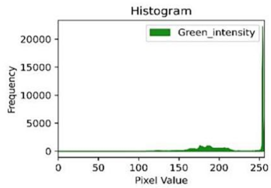
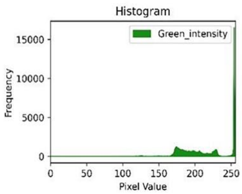
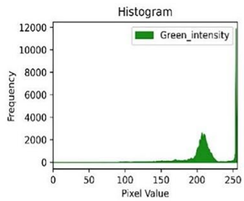
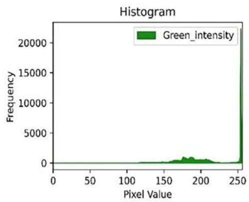
4.3 HQCM model performance evaluation on multi-nutrient stress identification

The effectiveness of the proposed method is assessed using performance measures, such as recall, accuracy, F1 score, and precision. Accuracy, a widely used metric, measures the

Table 3 Image pre-processing changes in various parameters

Original image			Gaussian			Mean			Median			
Mean	S. D	G I	Mean	S D	PSNR	Mean	S D	PSNR	Mean	S D	PSNR	GI
179.84	56.46	1.17	179.83	56.07	42.83	179.87	55.95	43.88	179.87	55.95	48.46	1.18
201.52	42.63	1.14	201.52	42.41	45.68	203.25	40.34	47.90	203.25	40.34	49.43	1.16
203.25	40.57	1.16	203.25	40.40	46.70	203.34	40.34	47.90	203.25	40.34	51.86	1.19
201.52	42.63	1.14	201.52	42.41	45.68	201.53	42.33	46.94	201.53	42.33	51.69	1.16
174.65	50.18	1.22	174.66	49.85	43.62	174.67	49.74	44.63	174.67	49.74	50.67	1.26
180.61	49.52	1.31	180.59	49.03	41.26	189.75	58.20	45.13	180.58	48.85	46.50	1.39
132.07	71.55	1.30	132.07	71.14	41.39	180.58	48.85	43.42	163.97	46.87	49.09	1.31
164.00	47.66	1.39	163.98	47.10	41.25	163.97	46.87	42.61	132.07	71.05	54.94	1.40
166.39	69.37	1.23	166.38	68.86	40.52	166.38	68.71	42.14	166.38	68.71	47.25	1.31
170.83	69.95	1.24	170.84	69.61	41.99	170.85	69.21	43.50	170.85	69.51	48.46	1.27

Table 4 Histogram changes observed in the pre-processed images

Input image 	Input image 	Input image 
Original histogram 	Original histogram 	Original histogram 
Histogram changes in after applying mean (3x3) filter 	Histogram changes in after applying mean (3x3) filter 	Histogram changes in after applying mean (3x3) filter 
Histogram changes in after applying median (3x3) filter 	Histogram changes in after applying median (3x3) filter 	Histogram changes in after applying median (3x3) filter 
Histogram changes in after applying Gaussian (3x3) filter 	Histogram changes in after applying Gaussian (3x3) filter 	Histogram changes in after applying Gaussian (3x3) filter 

model's capability to recognize multi-nutrient stress correctly. Precision quantifies the percentage of accurately recognized positive cases to the model's total number of positive predictions (TNPP). Recall evaluates the ratio of properly noticed positive cases to the dataset's TNPP. The F1 score, a comprehensive metric combining precision and recall, provides a balanced assessment of the overall efficacy of the model. The mathematical formulations delineating accuracy, precision, recall, and F1-score are elucidated through Eqs. (9) to (12).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{9}$$

$$Precision = \frac{TP}{TP + FP} \tag{10}$$

$$Recall = \frac{TP}{TP + FN} \tag{11}$$

$$F1 - score = \frac{2(precision * recall)}{precision + recall} \tag{12}$$

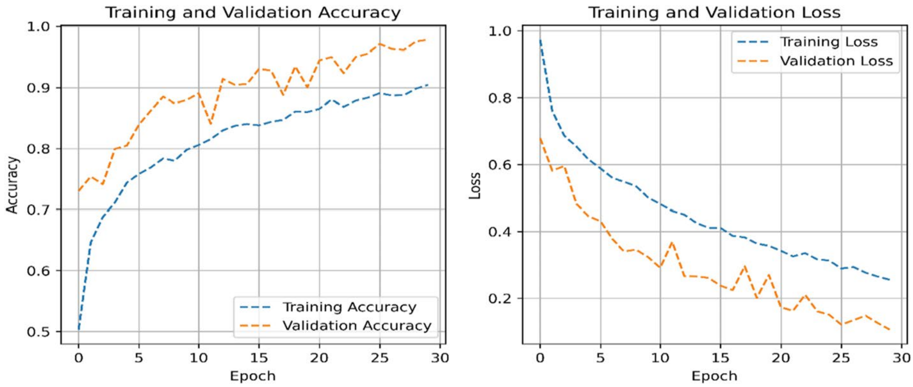
Here TP, TN, FP and FN specify the true positive, true negative, false positive, false negative, respectively. The experimental findings have been obtained using the performance metrics specified in Eqs. (9) to (12). As expounded in Sect. 3, the approach utilizes a solitary quantum convolutional layer combined with three classical convolutional layers. Traditional classification algorithms effectively manage large image dimensions without incurring excessive computing costs. In sharp contrast, quantum processes encounter constraints in handling high image dimensions, principally attributable to the dimensions of the encoding circuit and the reliance of the quantum mask on qubits. Moreover, the training procedure requires additional time and computing power due to the involvement of a single quantum layer and two quantum circuits in the image data processing. As a result, to enhance computational efficiency, we have chosen to train the proposed model with a reduced image size of 14×14.

Table 5 presents the experimental results, where three datasets are used for experimentation: multi-nutrient groundnut, rice plant, and plant village. Across all three datasets, the proposed method outperforms Xception [29], ResNet50V2 [30], DWC [49], DECM [50], and FQ_{Conv} [55] algorithms. Specifically, on the groundnut dataset, the proposed model achieves an accuracy of 97.79%, surpassing other algorithms. This superiority is attributed to the inherent quantum properties, efficient feature extraction via quantum circuits, and smaller image sizes. Similarly, the proposed method performs better on the rice and plant village datasets than existing algorithms. The training process entailed the utilization of 30 epochs, a learning rate set at 0.001, and a batch size of 16, with these hyperparameters being manually configured. Figure 6 displays the validation accuracy (VA), training accuracy (TA), validation loss (VL), and training loss (TL) graphs, indicating that the model neither overfits nor underfits but, the proposed model faces some difficulties in implementation in the real-time environment, such as quantum hardware constraints, coherence, gate fidelity, circuit depth, integration with the classical systems, quantum–classical communication, limited qubit availability, data gathering, and error accumulation.

The proposed method employs K-fold cross-validation (KCV), a commonly used classification methodology, to validate the obtained results. KCV divides the dataset into K subsets of equal size, and the model undergoes KCV, undergoing K training and testing cycles. During each cycle, one subset is earmarked for testing; K-1 subsets are used for training during the training process. This iterative process enhances the model generalization, reducing the risk of overfitting. The KCV outcomes are detailed in Table 6.

Table 5 HQCM model experimental results

Dataset	Type	Method	Input image size	Accuracy	Precision	Recall	F1-score	Parameters	Memory
Groundnut	Private	HQCM (Prop.)	14×14	97.79	97.78	97.78	97.78	93,231	364.18 KB
		Xception	224×224	85.43	84.29	85.19	85.39	3.4 M	13.07 MB
		ResNet50V2	224×224	82.79	82.46	81.73	82.02	23.8 M	91.04 MB
		DECM	224×224	82.14	82.17	81.86	82.17	2.8 M	10.68 MB
		DWC	224×224	92.31	90.25	91.27	85.89	14.7 M	56.42 MB
		FQ_{Conv}	20×20	93.72	92.17	93.04	91.04	142,539	509.13 KB
Rice plant	Public	HQCM (Prop.)	14×14	97.37	97.37	97.37	97.37	93,231	364.18 KB
		Xception	224×224	80.14	80.13	80.25	80.13	3.4 M	13.07 MB
		ResNet50V2	224×224	84.90	84.15	84.05	84.15	23.8 M	91.04 MB
		DECM	224×224	77.88	77.94	77.92	77.94	2.8 M	10.68 MB
		DWC	224×224	91.77	90.39	91.45	90.63	14.7 M	56.42 MB
		FQ_{Conv}	20×20	94.13	91.04	92.17	90.19	142,539	509.12 KB
Plant village	Public	HQCM (Prop.)	14×14	98.75	98.74	98.80	98.74	93,231	364.18 KB
		Xception	224×224	89.83	89.82	89.30	89.82	3.4 M	13.07 MB
		ResNet50V2	224×224	86.65	86.64	86.31	86.64	23.8 M	91.04 MB
		DECM	224×224	88.46	88.52	88.44	88.52	2.8 M	10.68 MB
		DWC	224×224	90.41	91.29	92.58	93.23	14.7 M	56.42 MB
		FQ_{Conv}	20×20	92.02	92.01	90.36	91.89	142,539	509.12 KB



(a) Proposed model training and VA

(b) Proposed model training and VL

Fig. 6 Proposed model performance analysis

4.4 NSLQ model performance evaluation on the multi-nutrient stress level quantification

HQCM and NSLQ (Nutrient Stress Level Quantification) are the two primary components of the proposed model. The HQCM technique is utilized to detect multi-nutrient stress in plants through the analysis of leaf coloring. In plant physiology, the NSLQ method quantitatively measures the multi-nutrient stress levels in plant leaves. This multi-nutrient stress level quantification is achieved primarily by evaluating the degree of green pigmentation in the leaves. The presence of green pigmentation in the leaves of plants is a reliable indicator of the plant's overall health status. The HQCM technique can detect and differentiate between different forms of multi-nutrient stress, including NP, PK, and KN. In addition, this module classifies plant leaf images into several folders based on nutrient stress categories, namely NP, PK, and KN. The folder images are divided into ten parts to examine multi-nutrient stress in plant leaves comprehensively. To measure the levels of multi-nutrient stress, we utilize the NSLQ method on the designated folders, utilizing the GI. The methodology for estimating the GI is elucidated in Sect. 3.3. Two datasets are employed to quantify the multi-nutrient stress, one on groundnut plants and the other on rice plants. The following sections will explore the experimental findings concerning quantifying multi-nutrient stress.

Table 6 Assessment of models using K-fold cross validation

Dataset	Type	Parameter	K = 1	K = 2	K = 3	K = 4	K = 5
Groundnut	Private	Accuracy	95.87	95.66	94.33	97.27	94.77
		Loss	12.10	10.70	15.21	07.48	14.99
Rice plant	Public	Accuracy	97.70	98.01	96.61	98.29	98.94
		Loss	06.52	05.55	08.85	04.46	08.02
Plant village	Public	Accuracy	99.48	99.22	98.96	98.33	99.87
		Loss	01.00	01.30	02.70	04.70	00.30

4.4.1 Groundnut dataset

This section provides further details regarding quantifying multi-nutrient stress in groundnut plants. Nutrient stress is when a plant lacks or excessively utilizes vital nutrients within a cultivated crop. Several observable symptoms characterize the manifestation of nutritional stress on plant leaves. For instance, the presence of yellowing leaves indicates nitrogen stress, while the emergence of brown and purple leaves suggests phosphorus and potassium nutrient stress, respectively. The existence of green pigmentation in the leaf of plants functions as a reliable sign of the overall physiological well-being of the plant. By harnessing the green color pigmentation observed in the leaves of plants, we can determine the existence of multi-nutrient stress in groundnut plants. A metric called the GI is employed to assess the level of green pigmentation, with a comprehensive explanation provided in Sect. 3.3. Table 7 displays the experimental findings about the GI in the groundnut plants. Using these GI values, we identified groundnut plants' nutrient stress levels in NP, PK, and KN.

The GI values are used to evaluate stress levels in groundnut crops. A GI value below 0.95 suggests the presence of low nutritional stress. On the contrary, a GI value surpassing 1.05 indicates high stress, but values within this range indicate the absence of nutrient stress. Table 7 displays the experimental findings on the GI values associated with the NK nutrients. The samples in this table are designated as S1 to S10. In the case of the S1 sample, the proposed model produced a GI value of 1.151, which exceeds the values produced by the NRAG (1.042) [43], CFM (0.986) [51], and DBS (0.863) [52] methods. The excellence of the proposed method originates from its utilization of PDF and mean value computations to calculate the GI, hence successfully eliminating image noise. As a result, the S1 sample is categorized as displaying high levels of nutrient stress in the crop, as the value of 1.151 surpasses the established threshold of 1.05. Likewise, samples S2 to S4 and S6 to S9 can be classified as exhibiting high nutritional stress in the cultivated plant. On the other hand, it is seen that samples S5 and S10 show low levels of nutritional stress in the cultivated plant. The nutrient stress levels for NP and PK samples are detailed in Tables 8 and 9, respectively.

4.4.2 Rice plant dataset

The experimental findings about nutritional stress levels in rice plants are shown in Tables 10, 11, and 12. These tables present the GI values of the KN, NP, and PK nutrients. The GI values show a substantial role in assessing the levels of multi-nutrient stress in rice plants. The HQCM model was utilized to identify plant stress by analyzing pigmentation in

Table 7 GI values comparison between proposed and existing methods on KN nutrient

Sample (→) Method (↓)	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10
NSLQ (Prop.)	1.151	1.152	1.185	1.207	0.726	1.232	1.230	1.186	1.204	0.807
NRAG	1.042	0.951	1.023	0.994	0.612	1.188	1.188	0.972	1.077	0.781
CFM	0.986	0.872	0.919	0.892	0.521	1.023	1.098	0.923	0.913	0.682
DBS	0.863	0.826	0.888	0.836	0.435	0.953	0.902	0.912	0.892	0.617

Table 8 GI values comparison between proposed and existing methods on NP nutrient

Sample (→) Method (↓)	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10
NSLQ (Prop.)	1.187	1.195	1.207	1.191	1.222	1.212	1.078	1.032	0.930	1.034
NRAG	0.950	1.072	1.043	1.038	0.955	1.069	1.008	0.993	0.880	0.915
CFM	0.912	0.983	1.021	1.001	0.893	1.002	1.000	0.983	0.817	0.892
DBS	0.843	0.943	0.975	0.943	0.823	0.932	0.987	0.924	0.791	0.839

Table 9 GI values comparison between proposed and existing methods on PK nutrient

Sample (→) Method (↓)	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10
NSLQ (Prop.)	1.179	1.217	1.210	1.170	1.120	1.066	1.045	1.094	1.210	1.133
NRAG	1.014	1.081	0.938	1.226	1.042	0.938	0.931	0.943	1.093	0.929
CFM	0.894	0.973	0.921	1.126	1.001	0.865	0.864	0.902	1.002	0.893
DBS	0.823	0.942	0.910	1.024	0.943	0.792	0.816	0.887	0.954	0.870

Table 10 GI values comparison between proposed and existing methods on KN nutrient

Sample (→) Method (↓)	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10
NSLQ (Prop.)	1.175	0.853	1.207	1.203	1.173	1.187	1.184	1.058	1.071	1.060
NRAG	1.076	0.791	1.029	1.021	1.048	0.986	1.059	0.989	0.983	0.978
CFM	1.001	0.729	0.973	0.952	0.973	0.899	1.004	0.977	0.969	0.932
DBS	0.983	0.632	0.895	0.899	0.915	0.817	0.943	0.912	0.928	0.917

Table 11 GI values comparison between proposed and existing methods on NP nutrient

Sample (→) Method (↓)	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10
NSLQ (Prop.)	1.215	1.240	1.214	1.193	1.211	1.178	1.183	1.077	1.072	0.941
NRAG	1.116	1.114	1.020	1.005	1.014	1.097	1.110	0.966	1.028	0.906
CFM	1.093	1.093	0.953	0.957	0.873	0.921	1.092	0.892	0.964	0.865
DBS	1.002	0.997	0.938	0.875	0.893	0.898	1.009	0.824	0.932	0.837

their leaves. The methodology employed in this study entailed organizing plant leaf images into separate folders to facilitate the identification of various levels of multi-nutrient stress.

The GI value is utilized to classify levels of multi-nutrient stress. The proposed method has shown superior performance compared to the NRAG [43], CFM [51], and DBS [52] algorithms due to the utilization of probability density function (PDF), mean values, and

Table 12 GI values comparison between proposed and existing methods on PK nutrient

Sample (→) Method (↓)	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10
NSLQ (Prop.)	1.190	1.186	1.234	1.2355	1.211	1.205	1.110	1.108	1.100	1.103
NRAG	0.985	1.116	1.083	0.9640	1.082	1.073	1.112	0.972	0.941	0.980
CFM	0.784	1.032	1.001	0.864	0.863	0.962	1.007	0.863	0.899	0.923
DBS	0.642	0.965	0.945	0.823	0.754	0.873	0.943	0.742	0.838	0.911

efficient noise reduction in the image. The GI values for nutritional stress in KN are presented in Table 10. Tables S1 to S10 correspond to the numerical identifiers assigned to the samples. In the instance of S1, the proposed method obtains a GI value of 1.175, surpassing the GI values of the NRAG (1.076), CFM (1.001), and DBS (0.983) algorithms. The given sample is classified as exhibiting a high level of multi-nutrient stress due to its GI value surpassing 1.05. In a similar vein, it can be observed that S2 exhibits a classification denoting a low amount of nutrient stress, whereas significant levels of nutrient stress characterize the remaining samples. The findings of NP and PK are displayed in Tables 11 and 12, respectively.

5 Conclusion

This research focuses on the examination of multi-nutrient stress in the leaves of plants. The model under consideration comprises two distinct modules, namely HQCM and NSLQ. The HQCM model is utilized for detecting multi-nutrient stress in plants, exhibiting a remarkable accuracy rate of 97.89%, surpassing the performance of current algorithms in this domain. This algorithm uses conventional and quantum computing technology, employing hybrid computing methodologies. The utilization of quantum computing is of paramount importance in the extraction of features from images via randomized quantum circuits. The aforementioned extracted features are utilized as inputs for a classifiable computing technology that is utilized in the process of image classification. The categorized images obtained are systematically arranged into several directories to facilitate later analysis of multi-nutrient stress levels. The NSLQ model uses stored folders to determine levels of multi-nutrient stress. The identification procedure uses green index (GI) values from plant leaves. The GI values play a significant role in classifying the diverse levels of multi-nutrient stress in plants. In the future, potential improvements for this study could entail expanding the proposed approach to tackle additional obstacles within the agricultural field, including disease diagnosis, weed identification, and nutrient recommendation systems.

Authors' contributions All the authors equally participated and contributed to bring the work to this shape. Their individual contributions are as provided below: *Kummari Venkatesh*: concepts, development of methodologies, Dataset gathering & creation, experimentation, analysis of result, and original draft writing; *K. Jairam Naik*: concepts, experimentation, results analysis, writing, document review, editing and entire supervision; *Achyut Shankar*: Document editing and overall supervision.

Funding It's not funded by any agency/organization either technically or financially. The work presented came from the PhD research at the Department of Computer Science and Engineering, The National Institute of Technology Raipur, Raipur, Chhattisgarh, India.

Data availability The data supporting the findings of this study can be obtained from the corresponding author upon request. However, the data is not publicly available, as privacy or ethical restrictions prevent its open dissemination.

Declarations

Ethical approval All authors have reviewed and approved the content of the manuscript, expressing their anticipation for the publication of this paper in the specified journal. We prioritize ethical use of plant imaging data by obtaining informed consent, anonymizing data, and ensuring robust security. Our practices comply with regulations, and we maintain transparency through stakeholder engagement. Periodic ethical reviews guide continuous improvement, addressing privacy concerns and upholding ethical standards.

Competing interests The authors assert that they have no identifiable conflicting financial or ethical interests.

References

1. Dunjko V, Briegel HJ (2018) Machine learning & artificial intelligence in the quantum domain: a review of recent progress. Rep Prog Phys 81(7):074001
2. Melnikov A, Kordzanganeh M, Alodjants A, Lee RK (2023) Quantum machine learning: From physics to software engineering. Adv Physics: X 8(1):2165452
3. Chen G, Long S, Yuan Z, Li W, Peng J (2023) Robustness and explainability of image classification based on QCNN. Quantum Eng 2023:1
4. Biamonte J, Wittek P, Pancotti N, Rebentrost P, Wiebe N, Lloyd S (2017) Quantum machine learning. Nature 549(7671):195–202
5. Schuld M, Fingerhuth M, Petruccione F (2017) Implementing a distance-based classifier with a quantum interference circuit. Europhys Lett 119(6):60002
6. Bokhan D, Mastiukova AS, Boev AS, Trubnikov DN, Fedorov AK (2022) Multiclass classification using quantum convolutional neural networks with hybrid quantum-classical learning. Frontiers in Physics 10:1069985
7. Lloyd S, Mohseni M, Rebentrost P (2013) Quantum algorithms for supervised and unsupervised machine learning. arXiv preprint https://doi.org/10.48550/arXiv.1307.0411
8. Shor PW (1994) Algorithms for quantum computation: discrete logarithms and factoring. In Proceedings 35th annual symposium on foundations of computer science (pp. 124–134). IEEE
9. Lloyd S (1996) Universal quantum simulators. Science 273(5278):1073–1078
10. Aaronson S, Chen L (2016) Complexity-theoretic foundations of quantum supremacy experiments. arXiv preprint https://doi.org/10.48550/arXiv.1612.05903
11. Wiebe N, Kapoor A, Svore KM (2014) Quantum deep learning. arXiv preprint https://doi.org/10.48550/arXiv.1412.3489
12. Havlíček V, Córcoles AD, Temme K, Harrow AW, Kandala A, Chow JM, Gambetta JM (2019) Supervised learning with quantum-enhanced feature spaces. Nature 567(7747):209–212
13. Harrow AW, Montanaro A (2017) Quantum computational supremacy. Nature 549(7671):203–209
14. Tang E (2019) A quantum-inspired classical algorithm for recommendation systems. In Proceedings of the 51st annual ACM SIGACT symposium on theory of computing (pp. 217–228)
15. Cong I, Choi S, Lukin MD (2019) Quantum convolutional neural networks. Nat Phys 15(12):1273–1278
16. Kossaifi J, Bulat A, Tzimiropoulos G, Pantic M (2019) T-net: Parametrizing fully convolutional nets with a single high-order tensor. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 7822–7831)
17. Henderson M, Shakya S, Pradhan S, Cook T (2020) Quanvolutional neural networks: powering image recognition with quantum circuits. Quantum Mach Intell 2(1):2
18. Broughton M, Verdon G, McCourt T, Martinez AJ, Yoo JH, Isakov SV, ... Mohseni M (2020) Tensorflow quantum: A software framework for quantum machine learning. arXiv preprint https://doi.org/10.48550/arXiv.2003.02989

19. Liu JG, Wang L (2018) Differentiable learning of quantum circuit born machines. Phys Rev A 98(6):062324
20. Wei S, Chen Y, Zhou Z, Long G (2022) A quantum convolutional neural network on NISQ devices. AAPPS Bull 32:1–11
21. Chen SYC, Wei TC, Zhang C, Yu H, Yoo S (2022) Quantum convolutional neural networks for high energy physics data analysis. Phys Rev Res 4(1):013231
22. Li Y, Zhou RG, Xu R, Luo J, Hu W (2020) A quantum deep convolutional neural network for image recognition. Quantum Sci Technol 5(4):044003
23. Preskill J (2018) Quantum computing in the NISQ era and beyond. Quantum 2:79
24. Aaronson S (2015) Read the fine print. Nature Phys 11:291–293. https://doi.org/10.1038/nphys3272
25. Goodfellow I, Bengio Y, Courville A (2016) Deep learning. MIT press
26. Yan F, Iliyasu AM, Le PQ (2017) Quantum image processing: a review of advances in its security technologies. Int J Quantum Inform 15(03):1730001
27. LeCun Y, Bengio Y, Hinton G (2015) Deep learning. Nature 521:436–444. https://doi.org/10.1038/nature14539
28. Mishra S, Tsai CY (2023) QSurfNet: a hybrid quantum convolutional neural network for surface defect recognition. Quantum Inf Process 22(5):179
29. Deva Priya VH, Juliet AV (2022) Automatic detection of Covid-19 based on xception network with optimized CNN. IETE J Res 1–9. https://doi.org/10.1080/03772063.2022.2138583
30. Büyükarıkan B, Ülker E (2023) Convolutional neural network-based apple images classification and image quality measurement by light colors using the color-balancing approach. Multimed Syst. https://doi.org/10.1007/s00530-023-01084-z
31. Cerezo M, Arrasmith A, Babbush R et al (2021) Variational quantum algorithms. Nat Rev Phys 3:625–644. https://doi.org/10.1038/s42254-021-00348-
32. Farhi E, Neven H (2018) Classification with quantum neural networks on near term processors. arXiv preprint https://doi.org/10.48550/arXiv.1802.06002
33. Schuld M, Bocharov A, Svore KM, Wiebe N (2020) Circuit-centric quantum classifiers. Phys Rev A 101(3):032308
34. Pesah A, Cerezo M, Wang S, Volkoff T, Sornborger AT, Coles PJ (2021) Absence of barren plateaus in quantum convolutional neural networks. Phys Rev X 11(4):041011
35. Qin Z, Lu X, Liu D, Nie X, Yin Y, Shen J, Loui AC (2023) Reformulating graph Kernels for self-supervised space-time correspondence learning. IEEE Trans Image Process. https://doi.org/10.1109/TIP.2023.3328485.
36. Yan L, Han C, Xu Z, Liu D, Wang Q (n.d.) Prompt learns prompt: exploring knowledge-aware generative prompt collaboration for video captioning. Proc Thirty-Second Int Joint Conf Artif Intell (IJCAI-23), (pp. 1622-1630)
37. Qin Z, Lu X, Nie X, Liu D, Yin Y, Wang W (2023) Coarse-to-fine video instance segmentation with factorized conditional appearance flows. IEEE/CAA J Autom Sin 10(5):1192–1208
38. Qin Z, Lu X, Nie X, Yin Y, Shen J (2023) Exposing the self-supervised space-time correspondence learning via graph kernels. Proc AAAI Conf Artif Intell 37(2):2110–2118
39. Liu D, Liang J, Geng T, Loui A, Zhou T (2023) Tripartite feature enhanced pyramid network for dense prediction. IEEE Trans Image Process. https://doi.org/10.1109/TIP.2023.3272826
40. Dunjko V, Taylor JM, Briegel HJ (2016) Quantum-enhanced machine learning. Phys Rev Lett 117(13):130501
41. Aïmeur E, Brassard G, Gambs S (2006) Machine learning in a quantum world. In Advances in Artificial Intelligence: 19th Conference of the Canadian Society for Computational Studies of Intelligence, Canadian AI 2006, Québec City, Québec, Canada, June 7-9, 2006. Proceedings 19 (pp. 431-442). Springer Berlin Heidelberg
42. Takeuchi Y, Morimae T, Hayashi M (2019) Quantum computational universality of hypergraph states with Pauli-X and Z basis measurements. Sci Rep 9(1):13585
43. Janani M, Jebakumar R (2023) Detection and classification of groundnut leaf nutrient level extraction in RGB images. Adv Eng Softw 175:103320
44. Bergholm V, Izaac J, Schuld M, Gogolin C, Ahmed S, Ajith V, ... Killoran N (2018) Pennylane: Automatic differentiation of hybrid quantum-classical computations. arXiv preprint https://doi.org/10.48550/arXiv.1811.04968
45. Venkatesh Kummari KJ (n.d.) Naik.: Groundnut Nutrient Deficiency Dataset. Accessed 02 May 2022. https://drive.google.com/file/d/1xNRX9gAlqMToWaJ9VbhIXh725dIcyCT/view?usp=sharing
46. Yu S, Xie L, Huang Q (2023) Inception convolutional vision transformers for plant disease identification. Internet Things 21:100650

47. Raksarikon W (n.d.) Nutrient deficiency symptom in rice, Kaggle V1. 2020. Available online: https://www.kaggle.com/guy007/nutrientdeficiencysymptomsinrice/activity (accessed on 7th March 2023)
48. OpenCV: Changing Colour Spaces. OpenCV. https://docs.opencv.org/4.x/df/d9d/tutorial py colorspaces.html Accessed 30 Sep 2022
49. Sunil GC, Zhang Y, Koparan C, Ahmed MR, Howatt K, Sun X (2022) Weed and crop species classification using computer vision and deep learning technologies in greenhouse conditions. J Agric Food Res 9:100325. https://doi.org/10.1016/j.jafr.2022.100325
50. Poyraz AK, Dogan S, Akbal E, Tuncer T (2022) Automated brain disease classification using exemplar deep features. Biomed Sig Process Control 73:103448. https://doi.org/10.1016/j.bspc.2021.103448
51. Sampathkumar S, Rajeswari R (2022) An automated crop and plant disease identification scheme using cognitive fuzzy C-means algorithm. IETE J Res 68(5):3786–3797. https://doi.org/10.1080/03772063.2020.1780163
52. Chen H, Liang M, Liu W, Wang W, Liu PX (2022) An approach to boundary detection for 3D point clouds based on DBSCAN clustering. Pattern Recogn 124:108431. https://doi.org/10.1016/j.patcog.2021.108431
53. Singh AL, Basu MS, Singh NB (2004) Mineral Disorders of groundnut. National Research Centre for groundnut (ICAR), Junagadh, India, p 85
54. Government of Tamilnadu (2020) Expert system for paddy, "Nutrient management". http://www.agritech.tnau.ac.in/expert_system/paddy/nutrientmanagement.html#disorders. Accessed 6 May 2023
55. Jing Y, Li X, Yang Y, Wu C, Fu W, Hu W, Xu H (2022) RGB image classification with quantum convolutional ansatz. Quantum Inf Process 21(3):101